

7.1.

Let $\{f_n: X \rightarrow Y\}$ be a uniformly convergent sequence of bounded functions.

Prove it is uniformly bounded.

Proof. Given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow \sup_x |f_n - f| \leq \epsilon$, where f is the limit function of f_n .

Therefore, $\sup_x |f| \leq \sup_x |f_N| + \epsilon$, the limit function f is bounded. Now,

for each $1 \leq i < N$, let M_i be a bound of f_i . Then, for all $k \in \mathbb{N}$,

$$\sup |f_k| \leq \max\{M_1, \dots, M_{N-1}, \sup |f| + \epsilon\}.$$

7.2-3. Let $f_n, g_n: X \rightarrow Y$ be functions.

1). If $\{f_n\}, \{g_n\}$ converges uniformly, then $\{f_n + g_n\}$ converges uniformly.

2). If $\{f_n\}, \{g_n\}$ converges uniformly of bounded functions, then $\{f_n g_n\}$ converges uniformly.

3). Find Counter example for 2), the boundedness omitted.

Proof. 1) can be shown directly, by $\sup |f_n + g_n| \leq \sup |f_n| + \sup |g_n|$.

But 2) is needed the boundedness condition for example,

$$f_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n}{n+1} \cdot \frac{1}{x} \quad \left(\sup |f_n - \frac{1}{x}| = \frac{1}{n+1} \rightarrow 0 \right)$$

$$g_n: [0, 1] \rightarrow \mathbb{R}: x \mapsto \frac{n+1}{n^2} \cdot \frac{1}{x} \quad \left(\sup |g_n - \frac{1}{x}| = \frac{1}{n} + \frac{1}{n^2} \rightarrow 0 \right)$$

But $f_n g_n(x) = \frac{1}{nx}$ is not uniformly convergent, because $\sup |f_n g_n - 0| = \infty$, but $f_n g_n \rightarrow 0$.

Now suppose that f_n and g_n are bounded (of course, uni. convergent.)

By the above problem, f_n and g_n are uniformly bounded. Now,

$$|f_n g_n - f g| = |(f_n - f) \cdot g_n + (g_n - g) \cdot f| \leq |g_n| |f_n - f| + |f| |g_n - g|$$

gives the result, (More specifically, given $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \sup |f_n - f|, \sup |g_n - g| < \epsilon.$$

Therefore, for these $n \geq N$,

$$\sup |f_n g_n - f g| \leq 2M\epsilon, \text{ where } M \text{ is bound of } |g_n| \text{ and } |f|.$$

7.4. Consider a map defined on Real Numbers

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2x}$$

- 1). Find $x \in \mathbb{R}$ for the series Converges absolutely
- 2) Find interval for the series Converges Uniformly
- 3) Find interval for the Converges Uniformly fails.
- 4). If the series converges, is f continuous? $\exists b \in \frac{1}{2}d \in \mathbb{R}$
- 5). " , f bounded?

Sol). Since $\left| \frac{1}{1+n^2x} \right| \leq \frac{1}{n^2c}$ if $x > 0$, for all $x > 0$, it Converges absolutely.

75. Let

$$f_n(x) = \begin{cases} 0 & x \notin \left[\frac{1}{n+1}, \frac{1}{n}\right] \\ \sin^2 \frac{\pi}{x} & x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \end{cases}$$

Show that $\{f_n\}$ converges to a continuous map, but not uniformly.

Observe that: $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right] \Leftrightarrow \frac{\pi}{x} \in [n\pi, (n+1)\pi]$

First, for any $x_0 \in \mathbb{R}$, $f_n(x_0) \rightarrow 0$ as $n \rightarrow \infty$. (If $x_0 \leq 0$, then clearly $f_n(x_0) = 0$, and if $x_0 > 0$, then for some $N \in \mathbb{N}$, $n \geq N \Rightarrow x_0 > \frac{1}{n} > \frac{1}{n+1} > 0$. Thus $f_n(x_0) = 0$, $\forall n \geq N$.)

But, for any $n \in \mathbb{N}$, $\sup |f_n(x) - 0| = 1 \neq 0$, thus it is not uniformly.

Meanwhile, see that

$$\sum_{n=1}^{\infty} f_n = \begin{cases} 0 & x \notin (0, 1] \\ \sin^2 \frac{\pi}{x} & x \in (0, 1] \end{cases}$$

Clearly, it converges absolutely, but not uniformly, because for all $N \in \mathbb{N}$,

$$\sup \left| \sum_{n=1}^{\infty} f_n - \sum_{i=1}^N f_i \right| = 1.$$

7.12. Let $g, f_n: (0, \infty) \rightarrow \mathbb{R}$ be Riemann-integrable on $[t, T]$ whenever $0 < t < T < \infty$.

If $|f_n| < g$, and $f_n \rightarrow f$ uniformly on every compact subset of $(0, \infty)$, and $\int_0^\infty g(x) dx < \infty$,

then
$$\lim_{n \rightarrow \infty} \int_0^\infty f_n(x) dx = \int_0^\infty \lim_{n \rightarrow \infty} f_n(x) dx.$$

Proof.

17. 20. If f is continuous on $[0,1]$ and

$$\int_0^1 f(x) \cdot x^n dx = 0 \quad \text{for all } n=0,1,\dots$$

Then $f=0$.

sol). By the Weierstrass Thm, there exists a sequence of polynomials $\{p_n\}$ s.t. $p_n \rightarrow f$.

And, since f and p_n are continuous on the compact set $[0,1]$, they are bounded.

Therefore $f \cdot p_n \rightarrow f \cdot f = f^2$. Now, clearly $f \cdot p_n \in \mathcal{P}$, thus

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \cdot p_n(x) dx = \int_0^1 \lim_{n \rightarrow \infty} f(x) p_n(x) dx = \int_0^1 [f(x)]^2 dx$$

But by the assumption, for each $n \in \mathbb{N}$, $\int_0^1 f(x) p_n(x) dx = 0$.

Therefore, $\int_0^1 [f(x)]^2 dx = 0$, thus $f^2 = 0$, by $f^2 \geq 0$. Consequently $f=0$.